SAFETY DATA SHEET

according to Regulation (EC) No. 1907/2006

Creation Date 05-Dec-2005 **Revision Date** 28-Jan-2024 **Revision Number** 3

SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description:	**2-Butyne-1,4-diol**
Cat No. :	**L06127**
Synonyms	1,4-Dihydroxy-2-butyne
Index No	603-076-00-9
CAS No	110-65-6
EC No	203-788-6
Molecular Formula	C4 H6 O2
REACH registration number	-

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use	Laboratory chemicals.
Sector of use	SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites
Product category	PC21 - Laboratory chemicals
Process categories	PROC15 - Use as a laboratory reagent
Environmental release category	ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates)
Uses advised against	No Information available

1.3. Details of the supplier of the safety data sheet

Company .

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US:**001-201-796-7100 / **Europe:** +32 14 57 52 99
CHEMTREC Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

SECTION 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Acute oral toxicity	Category 3 (H301)
Acute dermal toxicity	Category 3 (H311)
Acute Inhalation Toxicity - Dusts and Mists	Category 3 (H331)
Skin Corrosion/Irritation	Category 1 B (H314)
Serious Eye Damage/Eye Irritation	Category 1 (H318)
Skin Sensitization	Category 1 (H317)
Specific target organ toxicity - (single exposure)	Category 3 (H335)
Specific target organ toxicity - (repeated exposure)	Category 2 (H373)

Environmental hazards
Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements

Signal Word **Danger**

Hazard Statements

H314 - Causes severe skin burns and eye damage
H317 - May cause an allergic skin reaction
H335 - May cause respiratory irritation
H373 - May cause damage to organs through prolonged or repeated exposure
H301 + H311 + H331 - Toxic if swallowed, in contact with skin or if inhaled

Precautionary Statements

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing
P280 - Wear protective gloves/protective clothing/eye protection/face protection
P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing
P310 - Immediately call a POISON CENTER or doctor/physician
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Toxicity to Soil Dwelling Organisms
Toxic to terrestrial vertebrates
This product does not contain any known or suspected endocrine disruptors

SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1. Substances

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
1,4-Butynediol	110-65-6	EEC No. 203-788-6	>95	Acute Tox. 3 (H331) Acute Tox. 3 (H301) Acute Tox. 3 (H331) Skin Corr. 1B (H314) Eye Dam. 1 (H318) Skin Sens. 1 (H317) STOT RE 2 (H373) STOT SE 3 (H335)
Formaldehyde	50-00-0	200-001-8	0.01-0.1	Acute Tox. 3 (H301) Acute Tox. 3 (H311) Acute Tox. 3 (H331) Skin Corr. 1B (H314) Eye Dam. 1 (H318) Skin Sens. 1 (H317) STOT SE 3 (H335) Muta. 2 (H341) Carc. 1B (H350)

Component	Specific concentration limits (SCL's)	M-Factor	Component notes
1,4-Butynediol	Skin Corr. 1B :: C>=50% Skin Irrit. 2 :: 25%<=C<50% Eye Irrit. 2 :: 25%<=C<50%	-	-
Formaldehyde	Skin Corr. 1B :: C>=25% Eye Irrit. 2 :: 5%<=C<25% Skin Irrit. 2 :: 5%<=C<25% Skin Sens. 1 :: C>=0.2% STOT SE 3 :: C>=5%	-	-

REACH registration number	-	
Components	**Reach Registration Number**	
Formaldehyde	01-2119488953-20	
2-Butyne-1,4-diol	01-2119489899-05	

Full text of Hazard Statements: see section 16

SECTION 4: FIRST AID MEASURES

4.1. Description of first aid measures

General Advice Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.

Eye Contact Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

Skin Contact Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

Ingestion Do NOT induce vomiting. Call a physician or poison control center immediately.

Inhalation Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration.

Self-Protection of the First Aider Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. May cause allergic skin reaction. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician Treat symptomatically.

SECTION 5: FIREFIGHTING MEASURES

5.1. Extinguishing media

Suitable Extinguishing Media
Water spray, carbon dioxide (CO2), dry chemical, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons
No information available.

5.2. Special hazards arising from the substance or mixture

The product causes burns of eyes, skin and mucous membranes.

Hazardous Combustion Products
Carbon monoxide (CO), Carbon dioxide (CO_2), Thermal decomposition can lead to release of irritating gases and vapors.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Use personal protective equipment as required. Evacuate personnel to safe areas. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid dust formation.

6.2. Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information.

6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

SECTION 7: HANDLING AND STORAGE

7.1. Precautions for safe handling

Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not ingest. If swallowed then seek immediate medical assistance. Do not breathe (dust, vapor, mist, gas). Avoid dust formation.

Hygiene Measures
Handle in accordance with good industrial hygiene and safety practice.

7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK) (Germany) Storage Class/LGK 6.1C

Switzerland - Storage of hazardous substances Storage class - SC 6.1
https://www.kvu.ch/de/themen/stoffe-und-produkte
https://www.kvu.ch/fr/themes/substances-et-produits
https://www.kvu.ch/it/temi/sostanze-e-prodotti

7.3. Specific end use(s)

Use in laboratories

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1. Control parameters

Exposure limits
List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE -** 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

Component	European Union	The United Kingdom	France	Belgium	Spain
1,4-Butynediol	TWA: 0.5 mg/m³ (8h)	STEL: 1.5 mg/m³ 15 min	TWA / VME: 0.5 mg/m³ (8 heures).	TWA: 0.5 mg/m³ 8 uren	TWA / VLA-ED: 0.5 mg/m³ (8 horas)
Formaldehyde	TWA: 0.37 mg/m³ (8h) TWA: 0.3 ppm (8h) Skin STEL: 0.74 mg/m³ (8h) STEL: 0.6 ppm (8h)	STEL: 2 ppm 15 min STEL: 2.5 mg/m³ 15 min TWA: 2 ppm 8 hr TWA: 2.5 mg/m³ 8 hr Carc.	TWA / VME: 0.5 ppm (8 heures). for the healthcare, funeral and embalming sectors until July 11, 2024 TWA / VME: 0.3 ppm (8 heures). TWA / VME: 0.37 mg/m³ (8 heures). TWA / VME: 0.62 mg/m³ (8 heures). for the healthcare, funeral and embalming sectors until July 11, 2024 STEL / VLCT: 0.6 ppm. restrictive limit STEL / VLCT: 0.74 mg/m³. restrictive limit	STEL: 0.3 ppm 15 minuten STEL: 0.38 mg/m³ 15 minuten	STEL / VLA-EC: 0.6 ppm (15 minutos). STEL / VLA-EC: 0.74 mg/m³ (15 minutos). TWA / VLA-ED: 0.3 ppm (8 horas) TWA / VLA-ED: 0.37 mg/m³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
1,4-Butynediol	TWA: 0.5 mg/m³ 8 ore.	TWA: 0.36 mg/m³ (8	TWA: 0.5 mg/m³ 8 horas	TWA: 0.5 mg/m³ 8 uren	TWA: 0.14 ppm 8

	Time Weighted Average	Stunden). AGW - exposure factor 1 TWA: 0.1 ppm (8 Stunden). AGW - exposure factor 1 TWA: 0.1 ppm (8 Stunden). MAK TWA: 0.36 mg/m^3 (8 Stunden). MAK Höhepunkt: 0.1 ppm Höhepunkt: 0.36 mg/m^3 Haut			tunteina TWA: 0.5 mg/m^3 8 tunteina
Formaldehyde	TWA: 0.37 mg/m^3 8 ore. Time Weighted Average TWA: 0.3 ppm 8 ore. Time Weighted Average TWA: 0.62 mg/m^3 8 ore. Time Weighted Average for the health care, funeral and embalming sectors until July 11, 2024 TWA: 0.5 ppm 8 ore. Time Weighted Average for the health care, funeral and embalming sectors until July 11, 2024 STEL: 0.74 mg/m^3 15 minuti. Short-term STEL: 0.6 mg/m^3 15 minuti. Short-term Pelle	TWA: 0.3 ppm (8 Stunden). AGW - exposure factor 2 TWA: 0.37 mg/m^3 (8 Stunden). AGW - exposure factor 2 TWA: 0.3 ppm (8 Stunden). MAK no irritation should occur during mixed exposure TWA: 0.37 mg/m^3 (8 Stunden). MAK no irritation should occur during mixed exposure Höhepunkt: 0.6 ppm Höhepunkt: 0.74 mg/m^3	STEL: 0.6 ppm 15 minutos STEL: 0.74 mg/m^3 15 minutos Ceiling: 0.3 ppm TWA: 0.3 ppm 8 horas TWA: 0.37 mg/m^3 8 horas TWA: 0.62 mg/m^3 8 horas TWA: 0.5 ppm 8 horas	STEL: 0.5 mg/m^3 15 minuten TWA: 0.15 mg/m^3 8 uren	TWA: 0.3 ppm 8 tunteina TWA: 0.37 mg/m^3 8 tunteina STEL: 0.6 ppm 15 minuutteina STEL: 0.74 mg/m^3 15 minuutteina

Component	Austria	Denmark	Switzerland	Poland	Norway
1,4-Butynediol	MAK-TMW: 0.14 ppm 8 Stunden MAK-TMW: 0.5 mg/m^3 8 Stunden	TWA: 0.5 mg/m^3 8 timer STEL: 1 mg/m^3 15 minutter	Haut/Peau STEL: 0.36 mg/m^3 15 Minuten STEL: 0.1 ppm 15 Minuten TWA: 0.36 mg/m^3 8 Stunden TWA: 0.1 ppm 8 Stunden	STEL: 0.5 mg/m^3 15 minutach TWA: 0.25 mg/m^3 8 godzinach	TWA: 0.5 mg/m^3 8 timer STEL: 1.5 mg/m^3 15 minutter. value calculated
Formaldehyde	MAK-KZGW: 0.6 ppm 15 Minuten MAK-KZGW: 0.74 mg/m^3 15 Minuten MAK-TMW: 0.3 ppm 8 Stunden MAK-TMW: 0.37 mg/m^3 8 Stunden	TWA: 0.3 ppm 8 timer TWA: 0.37 mg/m^3 8 timer STEL: 0.74 mg/m^3 15 minutter STEL: 0.6 ppm 15 minutter	STEL: 0.6 ppm 15 Minuten STEL: 0.74 mg/m^3 15 Minuten TWA: 0.3 ppm 8 Stunden TWA: 0.37 mg/m^3 8 Stunden	STEL: 0.74 mg/m^3 15 minutach TWA: 0.37 mg/m^3 8 godzinach	TWA: 0.37 mg/m^3 8 timer TWA: 0.3 ppm 8 timer STEL: 0.74 mg/m^3 15 minutter. value from the regulation STEL: 0.6 ppm 15 minutter. value from the regulation Ceiling: 1 ppm Ceiling: 1.2 mg/m^3

Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
1,4-Butynediol	TWA: 0.5 mg/m^3	TWA-GVI: 0.5 mg/m^3 8 satima.	TWA: 0.5 mg/m^3 8 hr. STEL: 1.5 mg/m^3 15 min	TWA: 0.5 mg/m^3	TWA: 0.5 mg/m^3 8 hodinách. Potential for cutaneous absorption Ceiling: 1 mg/m^3
Formaldehyde	TWA: 0.37 mg/m^3 TWA: 0.3 ppm TWA: 0.62 mg/m^3 STEL : 0.5 ppm STEL : 0.74 mg/m^3 STEL : 0.6 ppm	TWA-GVI: 0.3 ppm 8 satima. except health, funeral and embalming sector TWA-GVI: 0.37 mg/m^3 8 satima. except health, funeral and embalming sector TWA-GVI: 0.5 ppm 8 satima. applies to health, funeral and	TWA: 0.3 ppm 8 hr. TWA: 0.5 ppm 8 hr. for the healthcare, funeral and embalming sectors until July 11, 2024 TWA: 0.37 mg/m^3 8 hr. TWA: 0.62 mg/m^3 8 hr. for the healthcare, funeral and embalming sectors until July 11, 2024	STEL: 0.74 mg/m^3 STEL: 0.6 ppm TWA: 0.3 ppm TWA: 0.37 mg/m^3	TWA: 0.37 mg/m^3 8 hodinách. Potential for cutaneous absorption Ceiling: 0.74 mg/m^3

		embalming sector applies until July 11, 2024 TWA-GVI: 0.62 mg/m³ 8 satima. applies to health, funeral and embalming sector applies until July 11, 2024 STEL-KGVI: 0.6 ppm 15 minutama. STEL-KGVI: 0.74 mg/m³ 15 minutama.	STEL: 0.6 ppm 15 min STEL: 0.738 mg/m³ 15 min STEL: 0.62 mg/m³ 15 min		

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
1,4-Butynediol	TWA: 0.5 mg/m³ 8 tundides.	TWA: 0.5 mg/m³ 8 hr	TWA: 0.5 mg/m³	TWA: 0.5 mg/m³ 8 órában. AK	TWA: 0.5 mg/m³ 8 klukkustundum. Ceiling: 1 mg/m³
Formaldehyde	TWA: 0.3 ppm 8 tundides. TWA: 0.37 mg/m³ 8 tundides. TWA: 0.62 mg/m³ 8 tundides. in the health, funeral and embalming sectors;valid until July 10, 2024 TWA: 0.5 ppm 8 tundides. in the health, funeral and embalming sectors;valid until July 10, 2024 STEL: 0.6 ppm 15 minutites. STEL: 0.74 mg/m³ 15 minutites.		STEL: 0.6 ppm STEL: 0.74 mg/m³ TWA: 0.3 ppm TWA: 0.37 mg/m³	STEL: 0.74 mg/m³ 15 percekben. CK TWA: 0.37 mg/m³ 8 órában. AK lehetséges borön keresztüli felszívódás	STEL: 0.6 ppm STEL: 0.74 mg/m³ TWA: 0.3 ppm 8 klukkustundum. TWA: 0.37 mg/m³ 8 klukkustundum. Skin notation

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
1,4-Butynediol	TWA: 0.5 mg/m³	TWA: 0.5 mg/m³ IPRD	TWA: 0.5 mg/m³ 8 Stunden	TWA: 0.5 mg/m³	TWA: 0.5 mg/m³ 8 ore
Formaldehyde	STEL: 0.74 mg/m³ STEL: 0.6 ppm TWA: 0.37 mg/m³ TWA: 0.62 mg/m³ TWA: 0.3 ppm TWA: 0.5 ppm	TWA: 0.3 ppm IPRD TWA: 0.37 mg/m³ IPRD TWA: 0.62 mg/m³ IPRD for healthcare, funeral, and embalming industries TWA: 0.5 ppm IPRD for healthcare, funeral, and embalming industries STEL: 0.74 mg/m³ STEL: 0.6 ppm			TWA: 1 ppm 8 ore TWA: 1.2 mg/m³ 8 ore STEL: 2 ppm 15 minute STEL: 3 mg/m³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
1,4-Butynediol	MAC: 1 mg/m³	TWA: 0.5 ppm	TWA: 0.5 mg/m³ 8 urah Koža STEL: 0.5 mg/m³ 15 minutah	TLV: 0.5 mg/m³ 8 timmar. NGV	
Formaldehyde	Skin notation MAC: 0.5 mg/m³	Ceiling: 0.74 mg/m³ TWA: 0.3 ppm TWA: 0.37 mg/m³	TWA: 0.62 mg/m³ 8 urah applies for health care, funeral and embalming activities until July 11, 2024 TWA: 0.5 ppm 8 urah applies for health care, funeral and embalming activities until July 11, 2024 TWA: 0.37 mg/m³ 8 urah TWA: 0.3 ppm 8 urah Koža STEL: 0.6 ppm 15	Binding STEL: 0.6 ppm 15 minuter Binding STEL: 0.74 mg/m³ 15 minuter TLV: 0.3 ppm 8 timmar. NGV TLV: 0.37 mg/m³ 8 timmar. NGV Hud	

			minutah STEL: 0.74 mg/m³ 15 minutah		

Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.
MDHS14/3 General methods for sampling and gravimetric analysis of respirable and inhalable dust

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

Workers; See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
1,4-Butynediol 110-65-6 (>95)		DNEL = 6.6mg/kg bw/day		DNEL = 0.2mg/kg bw/day
Formaldehyde 50-00-0 (0.01-0.1)			DNEL = 37µg/cm2	DNEL = 240mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
1,4-Butynediol 110-65-6 (>95)	DNEL = 1mg/m³	DNEL = 100mg/m³	DNEL = 0.5mg/m³	DNEL = 1.25mg/m³
Formaldehyde 50-00-0 (0.01-0.1)	DNEL = 0.75mg/m³		DNEL = 0.375mg/m³	DNEL = 9mg/m³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
1,4-Butynediol 110-65-6 (>95)	PNEC = 0.0155mg/L			PNEC = 134mg/L	PNEC = 0.05mg/kg soil dw
Formaldehyde 50-00-0 (0.01-0.1)	PNEC = 0.44mg/L	PNEC = 2.3mg/kg sediment dw	PNEC = 4.44mg/L	PNEC = 0.19mg/L	PNEC = 0.2mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
1,4-Butynediol 110-65-6 (>95)	PNEC = 0.00155mg/L				
Formaldehyde 50-00-0 (0.01-0.1)	PNEC = 0.44mg/L	PNEC = 2.3mg/kg sediment dw			

8.2. Exposure controls

Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection	Goggles (European standard - EN 166)
Hand Protection	Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Nitrile rubber Neoprene Natural rubber PVC	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection	Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection	When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly
Large scale/emergency use	Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced **Recommended Filter type:** Particulates filter conforming to EN 143
Small scale/Laboratory use	Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. **Recommended half mask:-** Particle filtering: EN149:2001 When RPE is used a face piece Fit Test should be conducted
Environmental exposure controls	Prevent product from entering drains.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1. Information on basic physical and chemical properties

Physical State	Solid	
Appearance	Beige	
Odor	No information available	
Odor Threshold	No data available	
Melting Point/Range	54 - 58 °C / 129.2 - 136.4 °F	
Softening Point	No data available	
Boiling Point/Range	238 °C / 460.4 °F	
Flammability (liquid)	Not applicable	Solid
Flammability (solid,gas)	No information available	
Explosion Limits	No data available	
Flash Point	152 °C / 305.6 °F	**Method -** No information available
Autoignition Temperature	410 °C / 770 °F	
Decomposition Temperature	> 150°C	
pH	6.4	100 g/L aq.sol
Viscosity	Not applicable	Solid
Water Solubility	2960 g/L (20°C)	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	**log Pow**	
1,4-Butynediol	-0.73	
Formaldehyde	-0.35	
Vapor Pressure	1.33 mbar @ 102 °C	
Density / Specific Gravity	1.200	
Bulk Density	No data available	
Vapor Density	Not applicable	Solid

Particle characteristics No data available

9.2. Other information

Molecular Formula C4 H6 O2
Molecular Weight 86.09
Evaporation Rate Not applicable - Solid

SECTION 10: STABILITY AND REACTIVITY

10.1. Reactivity

None known, based on information available

10.2. Chemical stability

Stable under normal conditions.

10.3. Possibility of hazardous reactions

Hazardous Polymerization Hazardous polymerization does not occur.
Hazardous Reactions None under normal processing.

10.4. Conditions to avoid

Incompatible products. Excess heat.

10.5. Incompatible materials

Strong oxidizing agents. Strong acids. Strong bases. Finely powdered metals. Acid anhydrides. Acid chlorides.

10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO_2). Thermal decomposition can lead to release of irritating gases and vapors.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;
Oral Category 3
Dermal Category 3
Inhalation Category 3

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
1,4-Butynediol	176 mg/kg (Rat female)	659 mg/kg (Rat)	0.69 mg/L/4h (Rat)
Formaldehyde	500 mg/kg (Rat)	LD50 = 270 mg/kg (Rabbit)	0.578 mg/L (Rat) 4 h

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;
Respiratory Based on available data, the classification criteria are not met
Skin Category 1

Component	Test method	Test species	Study result
Formaldehyde 50-00-0 (0.01-0.1)	Skin sensitization Test method Patch Test Respiratory sensitization in vitro	Man guinea pig	Sensitizer Sensitization

May cause sensitization by skin contact

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

Not mutagenic in AMES Test

(f) carcinogenicity; Based on available data, the classification criteria are not met

The table below indicates whether each agency has listed any ingredient as a carcinogen

Component	EU	UK	Germany	IARC
Formaldehyde	Carc Cat. 1B	Cat 3		Group 1

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Category 3

Results / Target organs Respiratory system.

(i) STOT-repeated exposure; Category 2

Target Organs Liver, Kidney, spleen.

(j) aspiration hazard; Not applicable
Solid

Symptoms / effects,both acute and delayed Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing.

11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

SECTION 12: ECOLOGICAL INFORMATION

12.1. Toxicity

Ecotoxicity effects Do not empty into drains. Contains a substance which is:. Harmful to aquatic organisms. The product contains following substances which are hazardous for the environment.

Component	Freshwater Fish	Water Flea	Freshwater Algae
1,4-Butynediol	49.3 - 58.3 mg/L LC50 96 h	26.8 mg/L EC50 = 48 h	430 mg/L EC50 = 96 h 480 mg/L EC50 = 72 h
Formaldehyde	Leuciscus idus: LC50 = 15 mg/L 96h	EC50 = 20 mg/L 96h EC50 = 2 mg/L 48h	EC50 (72h) = 4.89 mg/L (Desmodesmus subspicatus)

Component	Microtox	M-Factor
1,4-Butynediol	EC50 = 1343 mg/L 48 h EC50 = 2940 mg/L 17 h	

12.2. Persistence and degradability Readily biodegradable

Persistence Persistence is unlikely.

Component	Degradability
Formaldehyde 50-00-0 (0.01-0.1)	Readily biodegradable (OECD guideline 301A, 301C and 301D) under aerobic and anaerobic conditions.

Degradation in sewage treatment plant Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

12.3. Bioaccumulative potential Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
1,4-Butynediol	-0.73	3.16
Formaldehyde	-0.35	No data available

12.4. Mobility in soil The product is water soluble, and may spread in water systems . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

12.5. Results of PBT and vPvB assessment Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

12.6. Endocrine disrupting properties

Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects

Persistent Organic Pollutant This product does not contain any known or suspected substance

Ozone Depletion Potential This product does not contain any known or suspected substance

SECTION 13: DISPOSAL CONSIDERATIONS

13.1. Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

European Waste Catalogue (EWC) According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms.

Switzerland - Waste Ordinance Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
https://www.fedlex.admin.ch/eli/cc/2015/891/en

SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

14.1. UN number UN2716
14.2. UN proper shipping name 1,4-Butynediol
14.3. Transport hazard class(es) 6.1
14.4. Packing group III

ADR

14.1. UN number	UN2716
14.2. UN proper shipping name	1,4-Butynediol
14.3. Transport hazard class(es)	6.1
14.4. Packing group	III

IATA

14.1. UN number	UN2716
14.2. UN proper shipping name	1,4-Butynediol
14.3. Transport hazard class(es)	6.1
14.4. Packing group	III

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

SECTION 15: REGULATORY INFORMATION

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories
Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
1,4-Butynediol	110-65-6	203-788-6	-	-	X	X	X	X	X
Formaldehyde	50-00-0	200-001-8	-	-	X	X	KE-17074	X	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
1,4-Butynediol	110-65-6	X	ACTIVE	X	-	X	X	X
Formaldehyde	50-00-0	X	ACTIVE	X	-	X	X	X

Legend: X - Listed '-' - Not Listed **KECL** - NIER number or KE number (http://ncis.nier.go.kr/en/main.do)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
1,4-Butynediol	110-65-6	-	Use restricted. See item 75. (see link for restriction details)	-
Formaldehyde	50-00-0	-	Use restricted. See item 72. (see link for restriction details) Use restricted. See item 28. (see link for restriction details) Use restricted. See item 75. (see link for restriction	-

			details)	

REACH links
https://echa.europa.eu/substances-restricted-under-reach

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
1,4-Butynediol	110-65-6	Not applicable	Not applicable
Formaldehyde	50-00-0	5 tonne	50 tonne

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals
Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?
Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .
Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values
Take note of Dir 76/769/EEC relating to restrictions on the marketing and use of certain dangerous substances and preparations

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification See table for values

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
1,4-Butynediol	WGK 2	Class I : 20 mg/m³ (Massenkonzentration)
Formaldehyde	WGK 3	Krebserzeugende Stoffe - : 5 mg/m³ (Massenkonzentration)

Component	France - INRS (Tables of occupational diseases)
Formaldehyde	Tableaux des maladies professionnelles (TMP) - RG 43

Swiss Regulations
Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).
Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Formaldehyde 50-00-0 (0.01-0.1)		Group I	

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

SECTION 16: OTHER INFORMATION

Full text of H-Statements referred to under sections 2 and 3
H301 - Toxic if swallowed
H311 - Toxic in contact with skin
H331 - Toxic if inhaled

H314 - Causes severe skin burns and eye damage
H317 - May cause an allergic skin reaction
H318 - Causes serious eye damage
H335 - May cause respiratory irritation
H373 - May cause damage to organs through prolonged or repeated exposure
H341 - Suspected of causing genetic defects
H350 - May cause cancer

Legend

CAS - Chemical Abstracts Service
EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances
PICCS - Philippines Inventory of Chemicals and Chemical Substances
IECSC - Chinese Inventory of Existing Chemical Substances
KECL - Korean Existing and Evaluated Chemical Substances
TSCA - United States Toxic Substances Control Act Section 8(b) Inventory
DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List
ENCS - Japanese Existing and New Chemical Substances
AICS - Australian Inventory of Chemical Substances
NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit
ACGIH - American Conference of Governmental Industrial Hygienists
DNEL - Derived No Effect Level
RPE - Respiratory Protective Equipment
LC50 - Lethal Concentration 50%
NOEC - No Observed Effect Concentration
PBT - Persistent, Bioaccumulative, Toxic
TWA - Time Weighted Average
IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)
LD50 - Lethal Dose 50%
EC50 - Effective Concentration 50%
POW - Partition coefficient Octanol:Water
vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road
IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code
OECD - Organisation for Economic Co-operation and Development
BCF - Bioconcentration factor
ICAO/IATA - International Civil Aviation Organization/International Air Transport Association
MARPOL - International Convention for the Prevention of Pollution from Ships
ATE - Acute Toxicity Estimate
VOC - (volatile organic compound)

Key literature references and sources for data
https://echa.europa.eu/information-on-chemicals
Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards	On basis of test data
Health Hazards	Calculation method
Environmental hazards	Calculation method

Training Advice
Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.
Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.
First aid for chemical exposure, including the use of eye wash and safety showers.

Prepared By	Health, Safety and Environmental Department
Creation Date	05-Dec-2005
Revision Date	28-Jan-2024
Revision Summary	New emergency telephone response service provider.

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006. COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No 1907/2006 .

For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2, Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and Preparations).

Disclaimer
The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information

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End of Safety Data Sheet